PORTFOLIO PROJECT · DELIVERABLES SUMMARY

AI and the Rural Care Gap

Compiled October 25, 2026 · Prepared for Neha Vittal

Project Overview

An interactive research microsite examining where artificial intelligence is measurably closing — and where it is not yet closing — gaps in rural healthcare access in the United States. The project grades 8 clinical AI interventions (diabetic eye screening, stroke triage, ambient documentation, obstetric ultrasound AI, and others) against peer-reviewed evidence, maps the rural–urban AI adoption gap using federal and industry data, documents known AI failure modes, and visualizes the $50B Rural Health Transformation Program that is funding this transition through FY2030. The site is built as a single self-contained, dependency-free web page with custom data visualizations and a sourced bibliography of 29 citations.

LIVE INTERACTIVE SITE

https://claude.ai/artifact/QYPoidJiAsyBYxinm9Zoio

Key Stats & Figures

29
primary sources cited (CDC, HRSA, FCC, CMS, NEJM, JAMIA, The Lancet, and others)

8
AI clinical use cases graded for evidence strength

6
rural care-access challenge categories covered

2
custom SVG data visualizations, built with no charting library

5
distinct AI failure modes identified and documented

4-pt
evidence-strength rating scale applied across all use cases

$50B
Rural Health Transformation Program funding, FY2026–2030 ($10B/yr)

$147M–$281M
per-state RHTP award range analyzed

3
audience-specific recommendation tracks (state offices, hospital leaders, researchers)

Resume Bullet Points

- Designed and built an interactive data journalism site analyzing AI's impact on rural healthcare access, synthesizing findings from 29 peer-reviewed studies and federal sources (CDC, HRSA, FCC, CMS) into a graded evidence framework across 8 clinical AI use cases
- Built 2 custom SVG data visualizations from scratch (no charting library) — a comparative mortality chart and a 3-metric toggleable adoption-gap chart — validated for colorblind accessibility using OKLab ΔE contrast testing across light and dark themes
- Analyzed a $50B federal funding program (Rural Health Transformation Program, $10B/year over 5 years, FY2026–2030) including state allocation caps and a $147M–$281M per-state award range, visualized as a live policy timeline
- Designed a 4-point evidence-strength rating system to grade AI healthcare interventions by rigor, applied consistently across 8 use cases spanning diagnostics, documentation, and triage
- Identified and documented 5 distinct AI failure modes in rural healthcare deployment, translating technical risk findings into plain-language guidance for non-technical stakeholders

- Shipped a fully responsive, dual-theme (light/dark) web application with zero external chart dependencies, built and published as a single self-contained deployable artifact

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